The Saka Law: Measuring Recursive Technological Acceleration and Its Implications for Biomedical Longevity

Javier Carrasco (“Saka”)

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Abstract

Technological forecasting often extrapolates individual trends—compute, biotechnology, energy, robotics or launch cost—without explicitly representing interactions between them.

This paper proposes the **Saka Law of Recursive Technological Acceleration**, a falsifiable forecasting hypothesis: when a technology materially improves the process by which new technology is discovered, validated and deployed, the *growth rate* of technological progress can itself increase.

The idea is not new in economics: it is the feedback parameter of the Romer–Jones idea-production function, and it underlies both the "accelerating returns" literature and the recent literature on AI-driven explosive growth. Existing evidence also points the other way: research productivity has been falling across many fields for decades. The contribution of this paper is therefore not the idea itself but a **multi-domain operationalization and a preregistered calibration protocol** for testing it.

The paper introduces the **Future Technology Acceleration Framework (FTAF)**, a systems model combining artificial intelligence, compute, energy, robotics, biotechnology, measurement, space access, manufacturing and clinical translation. It further introduces a Historical Technology Acceleration Baseline (HTAB), a Technology Acceleration Ratio (TAR), an Evidence Maturity Score (EMS), a Longevity Translation Index (LTI), a Bottleneck & Risk Index (BPI), a Diffusion & Accessibility Index (DAI), and an operational healthy-longevity derivative termed the LEV proxy.

The framework explicitly rejects indefinite exponential extrapolation. It compares linear, exponential, power-law, logistic and change-point models, incorporates physical and translational bottlenecks, and proposes Monte Carlo forecasting with calibration against future observations. It ends with dated, quantitative predictions that can fail.

A motivating application is biomedical longevity. The key question is not whether ageing will be "solved" by a particular date, but whether scientific and medical progress can increasingly offset age-related functional decline and allow individuals to reach successive generations of therapies. This dynamic is formalized as a **Technology Survival Ladder**.

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1. Introduction

Between the mid-1980s and the mid-2020s, computing, genomic measurement, communications, energy technologies, robotics and space launch systems changed by orders of magnitude. Forecasting the next decades by simply repeating historical averages, however, may be inadequate if the process that creates technology is itself becoming technologically amplified.

In 2026 several signals motivate examining this possibility. Frontier AI training compute continues to rise rapidly; algorithmic efficiency is improving; the stock of AI compute is expanding; autonomous task horizons are being measured over progressively longer tasks; self-driving laboratories are evolving from narrow automation toward systems that can propose, execute and interpret experiments; and AI systems are beginning to contribute to frontier mathematical research.

For example, on 8 September 2026 OpenAI published an AI-generated proof of finite-time blowup (singularity formation) for the three-dimensional incompressible Navier–Stokes equations, addressing the Millennium Prize Problem, together with a Lean formalization [1, 2]. The result concerns the **forced** problem: a smooth solution starting from rest and driven by a smooth external force. The official Clay formulation accepts such a breakdown as one of its four alternative resolutions, but the behaviour of the unforced equations remains open. At the time of writing, human experts still need to confirm that the formalized statement is equivalent to the problem as posed, and the Clay Mathematics Institute has not awarded the prize. It is best treated as evidence of a new research-capability regime, not as proof that all scientific problems can now be rapidly solved.

It is not an isolated case. In August 2026 a Claude model raised the proven lower bound on the proportion of Riemann zeta zeros on the critical line from 41.6% to 67.2%, with a Lean formalization [27]; in September 2026 another produced a complete formalization of the proof of Fermat's Last Theorem [28], which verifies known mathematics rather than discovering new results. Mathematics is, however, the domain in which AI can work without physical validation; Section 9 explains why such results are counted separately from the experimental sciences.

The central thesis of this paper is therefore modest but consequential:

> **If technology improves the rate at which new technology is generated, then technological progress can enter a recursively accelerated regime, subject to physical, biological, economic and institutional bottlenecks.**

This statement is termed the **Saka Law** for shorthand. It is proposed as a hypothesis to be measured and potentially falsified, not as a universal law of nature. Section 4 restates it in a form that can be estimated.

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2. Related work

Endogenous and semi-endogenous growth. Romer [3] made the production of ideas endogenous to the economy. Jones [4] wrote the idea-production function as

$$\dot{A} = \delta \, R^{\lambda} A^{\phi},$$

where A is the stock of knowledge, R research effort and ϕ the strength of the feedback from existing knowledge to the productivity of research. The Saka Law is, in this language, a claim about ϕ and about how technology now changes R itself. Kremer [5] used a closely related model to explain the historically accelerating growth of world population and output.

Ideas getting harder to find. Bloom, Jones, Van Reenen and Webb [6] found that research productivity has fallen steadily across semiconductors, agriculture, medicine and firm-level data—roughly 5% per year in aggregate—so that sustaining constant growth has required ever more researchers. This is the main empirical counterweight to any acceleration hypothesis and is the baseline the framework must beat.

Eroom's law. In drug development, the number of new drugs approved per billion dollars of R&D spending halved roughly every nine years between 1950 and 2010 [7], despite large gains in underlying technologies such as sequencing and high-throughput screening. This is directly relevant to the clinical-translation component L and to the longevity application.

Accelerating returns and explosive growth. Kurzweil's "Law of Accelerating Returns" [8] made an informal version of the recursive claim. Aghion, Jones and Jones [9] modelled AI as automation of tasks in both goods and idea production and showed that Baumol-type bottlenecks—tasks that remain essential but hard to automate—can prevent explosive growth even when most tasks are automated. Nordhaus [10] proposed empirical tests for an approaching "economic singularity" and found that most did not yet indicate one. Davidson [11] and Erdil and Besiroglu [12] reviewed the arguments for and against AI-driven explosive growth.

Forecasting technological progress. Nagy et al. [13] and Farmer and Lafond [14] compared Moore's-law-type and Wright's-law-type models on large sets of historical cost curves and showed how to produce calibrated forecast intervals from them. The HTAB and TAR constructs below follow this approach.

Position of this paper. The framework does not claim a new mechanism. Its contribution is to (i) track the feedback across several physical and institutional subsystems at once rather than in aggregate output alone, (ii) separate discovery from clinical translation explicitly, and (iii) commit to quantitative, dated predictions and a preregistered update procedure.

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3. Why recursive acceleration is plausible in 2026 — and why it might not be

3.1 AI resources are growing rapidly

Epoch AI reports that frontier language-model training compute has grown at roughly 5× per year since 2020, while the total stock of AI compute has grown around 3.4× per year since 2022. The same source estimates pre-training compute efficiency improving by roughly 3× per year [15].

The cost of reaching a *fixed* level of performance on mathematical and scientific reasoning benchmarks fell by about 47% per quarter (roughly 13× per year) from 2023, which Epoch AI reports as about six times faster than the fall in the cost of compute [26]. Section 6 explains why such a comparison between a three-year-old technology and decades-old ones overstates the contrast.

These variables should not be interpreted as equivalent to intelligence. They are inputs into an innovation process.

3.2 Autonomy is becoming measurable

METR's task-completion time-horizon methodology measures the duration of tasks, in human expert time, that frontier agents can complete at specified success probabilities. It provides a more behaviourally meaningful metric than raw parameter count and allows longitudinal tracking of agent autonomy [16].

3.3 Autonomous laboratories close part of the physical loop

Self-driving laboratories combine robotics, experimental platforms and AI. A 2026 *Nature Reviews Chemistry* review describes a transition from narrowly focused automation to multipurpose systems in which algorithms can propose, execute and interpret experiments with limited human intervention [17].

This matters because purely computational intelligence cannot fully accelerate experimental science if physical validation remains serial and human-limited.

3.4 Energy is a hard constraint

The International Energy Agency projects global data-centre electricity demand to roughly double from about 485 TWh in 2025 to about 950 TWh in 2030 in its 2026 update, with consumption from AI-focused data centres roughly tripling over the same period [18].

Thus, abundant compute cannot be modelled independently of energy infrastructure.

3.5 Biomedical translation is entering new regimes

In June 2026 the first participant was dosed in a Phase I trial of ER-100, an experimental partial epigenetic reprogramming therapy using controlled OCT4, SOX2 and KLF4 expression, in patients with glaucoma or non-arteritic anterior ischemic optic neuropathy (NAION) [19,

20]. The trial primarily evaluates safety and tolerability, so it must not be interpreted as evidence that systemic rejuvenation has been achieved.

Its importance to this framework is different: a class of interventions previously discussed mainly at preclinical level has crossed into human testing. This creates a measurable translation milestone.

3.6 Evidence against

A framework built only from supportive signals would be selected to confirm itself. The following observations point the other way and must be treated as part of the baseline:

- **Falling research productivity.** Across many domains, more research effort has been needed per unit of progress for decades [6].
- **Eroom's law.** Drug R&D output per dollar declined for sixty years despite large technological gains in the tools of discovery [7].
- **Low clinical success rates.** Only a minority of programmes entering Phase I reach approval—roughly one in seven across therapeutic areas, and far fewer in oncology [21]. Faster discovery that feeds the same attrition funnel does not by itself accelerate medicine.
- **Biological timescales.** Trials whose endpoints are morbidity or mortality in ageing populations take years by construction. Computation does not shorten the time it takes to observe that people live longer.
- **Slow longevity trend.** Record period life expectancy has risen by about 0.25 years per calendar year for more than 160 years [22]. Any claim of acceleration in healthy lifespan must show a departure from this remarkably stable line.
- **Automation bottlenecks.** In task-based models, the tasks that resist automation come to dominate cost and time, which can cap growth even when most tasks are automated [9].

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4. The Saka Law, stated formally

Let $T(t)$ denote a broad technological-capability state, and let $g(t) = \dfrac{d \ln T}{dt}$ be its growth rate.

Conventional progress can be represented as

$$\frac{dT}{dt} = f(T, R),$$

where R represents research resources. Recursive technological acceleration appears when part of T increases the productivity of the process producing future T:

$$\frac{dT}{dt} = f\big(T, R, A(T)\big),$$

where $A(T)$ is technological augmentation of discovery itself.

A positive second derivative, $\ddot{T} > 0$, is **not** sufficient evidence: any ordinary exponential trend satisfies it. The signature of recursive acceleration is a rising growth rate—super-exponential growth:

$$\frac{dg}{dt} = \frac{d^2 \ln T}{dt^2} > 0,$$

sustained while technology is materially increasing the productivity of research, design, experimentation, verification, manufacturing or deployment.

In the idea-production form of Section 2, the growth rate of knowledge is $g_A = \dot{A}/A = \delta R^{\lambda} A^{\phi-1}$. Following Bloom et al. [6], **research productivity** is growth per unit of effective research input:

$$\pi = \frac{g_A}{R^{\lambda}} = \delta A^{-\beta}, \qquad \beta = 1 - \phi,$$

with $\lambda = 1$ in the baseline of [6]; the value of λ is fixed at preregistration. Three regimes follow:

- $\beta > 0$ ($\phi < 1$): productivity falls as knowledge accumulates, and constant growth requires ever more research input. This is what [6] reports, with $\beta \approx 0.2$ for Moore's law and $\beta \approx 3$ for the aggregate economy.
- $\beta = 0$ ($\phi = 1$): productivity is constant, and constant research input gives constant exponential growth.
- $\beta < 0$ ($\phi > 1$): productivity rises with the stock of knowledge, and the growth rate rises even at constant research input. This is the recursive regime.

Two quantities must not be confused with π. First, idea output per unit of input, $\dot{A}/R^{\lambda} = \delta A^{\phi}$, rises whenever $\phi > 0$; it has been rising in semiconductors throughout the period in which π fell, and is not evidence of recursion. Second, the growth rate itself obeys

$$\frac{\dot{g}_A}{g_A} = \lambda\, g_R - \beta\, g_A,$$

so g_A can rise with $\beta > 0$ if the growth of research input accelerates. That is an investment boom, not recursion (Section 7.4). The Saka Law can therefore be stated as an estimable claim:

> **Recursive acceleration holds in a domain when research productivity π, measured against total research input, stops declining and begins to rise—equivalently, when the effective β, re-estimated on rolling windows, turns negative ($\phi > 1$).**

The sign change need not come from a change in ϕ as a deep parameter. In task-based models [9], automation lets capital and compute—themselves products of technology—supply part of effective research input, so that the cost of a unit of research falls. Measured against expenditure, as below, that appears as rising π.

Measuring R when AI is a research input. This condition is only testable if R is defined so that automation cannot inflate measured productivity by construction. If AI systems replace researchers and R counts only people, productivity rises mechanically even when nothing about the discovery process has improved. The framework therefore follows Bloom et al. [6], who measure effective research input as deflated R&D *expenditure*, and requires that expenditure to include compute, cloud services, model access and laboratory automation bought for research. Two variants are reported:

- R^{total} **(primary):** deflated research expenditure including all automated inputs. Only this variant counts as evidence for the hypothesis.

- R^{human} **(secondary):** research personnel only. It is reported to show the extent of substitution, but a rise in productivity measured against R^{human} alone is **not** evidence of recursive acceleration.

The deflator and the classification of expenditure lines are fixed at preregistration.

This condition is not expected to hold indefinitely. Energy, fabrication capacity, experiment duration, regulation, biological timescales, capital and safety constraints can reduce or reverse acceleration.

Accordingly, the stronger form of the Saka Law is not "technology is exponential". It is:

> **The growth regime of technology depends partly on the growth rate of the systems that generate validated technological knowledge.**

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5. Historical Technology Acceleration Baseline (HTAB)

A forecasting system should first ask whether the present regime is actually unusual relative to history.

For historical metric i measured at times $t_1 < t_2$, define its average log-growth rate:

$$k_i = \frac{\ln\left(V_{i,t_2}/V_{i,t_1}\right)}{t_2 - t_1}.$$

For declining-cost technologies, the ratio is inverted. In practice k_i should be estimated by regression on the full series rather than from two endpoints, with a standard error.

Potential HTAB series include:

- semiconductor density and compute;
- supercomputer performance;
- genomic sequencing cost;
- battery cost per kWh;
- photovoltaic module cost per watt;
- telecommunications bandwidth and cost;
- industrial robot density;
- launch cost and annual mass to orbit;
- power-electronics cost per watt and conversion efficiency (silicon, SiC, GaN);
- cost of a fixed level of AI performance [26].

Selection bias. Every series above is one that is already known to have improved dramatically. A baseline built only from successes will overstate historical growth in some respects and hide stagnation in others. The HTAB should therefore also include domains that slowed or stalled, for example new drugs approved per unit of R&D spending [7], crop yields, transport speed, and construction productivity. The weights w_i and the list of series must be fixed before current data are compared with them.

The historical baseline is a weighted log-growth average:

$$k_{HTAB} = \sum_i w_i \, k_i,$$

which is not a claim that the entire economy improves at one rate. Its purpose is to create a null model:

$$H_0 : k_{current} = k_{historical},$$

against which a modern acceleration regime can be tested.

Maturity. Most technologies improve fastest early in their life: cost falls with cumulative production (Wright's law [29]), and the first doublings of production come quickly. DNA sequencing, for example, fell much faster than Moore's law in the first years of next-generation sequencing. A comparison between a young technology and mature ones therefore mixes two effects: a possible change of regime, and an ordinary early-stage learning curve. For each HTAB series the framework records its **introduction date** and, where available, its **cumulative production**, so that series can also be compared at the same stage of maturity (Section 6).

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6. Technology Acceleration Ratio (TAR)

TAR is defined at two levels. For an individual HTAB domain i, compare its current log-growth rate with its *own* history:

$$TAR_i(t) = \frac{k_{i,current}(t)}{k_{i,historical}}.$$

For the aggregate, compare the weighted current rate with the weighted baseline:

$$TAR(t) = \frac{\sum_i w_i \, k_{i,current}(t)}{k_{HTAB}}.$$

Domain-level statements (such as prediction P1) use TAR_i; the aggregate TAR summarizes the whole set. Comparing one domain's current rate with the aggregate baseline would confuse "this domain has always been fast" with "this domain has accelerated".

For domains whose historical rate is near zero or negative (for example, stagnating series included to avoid selection bias), the ratio is unstable. For those, the difference $k_{i,current} - k_{i,historical}$ is reported instead.

Interpretation:

- $TAR < 1$: technological growth slower than the historical baseline;
- $TAR \approx 1$: similar to the historical baseline;
- $TAR > 1$: accelerated relative to baseline.

Calendar-aligned and maturity-aligned TAR. The ratios above are *calendar-aligned*: each series is compared with its own past. For a technology too young to have a past of its own, the comparison is instead *maturity-aligned*: its current log-growth rate is divided by the rates of the other HTAB series over the **same number of years since their introduction**, or over the same multiples of cumulative production where those data exist:

$$TAR_i^{mat}(t) = \frac{k_{i,current}(t)}{\bar{k}_{HTAB}(a_i)},$$

where a_i is the age of technology i and $\bar{k}_{HTAB}(a)$ the weighted average log-growth rate of the HTAB series at age a. Both forms are reported. A claim that a young technology has

entered an accelerated regime requires $TAR_i^{mat} > 1$, not only a high calendar-aligned rate. The fall in the cost of AI capability [26], with three years of data, is the motivating case.

Because $k_{current}$ is estimated on short windows, it is noisy. TAR must always be reported with an uncertainty interval, and a claim of acceleration requires the lower bound of that interval to exceed 1. A sustained $TAR > 1$ across independent domains would be much stronger evidence than a short-lived burst in a single field.

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7. Future Technology Acceleration Framework (FTAF)

7.1 State vector

The proposed state vector is

$$X_t = [A, C, E, R, B, D, S, M, L]$$

with each component normalized to its 2026 value (2026 = 1):

Symbol	Component	Candidate operational proxy
A	AI and scientific-reasoning capability	Composite index of sub-measures (see below); METR 50%-success task horizon [16] is one component
C	Compute availability	Installed AI compute stock, FLOP/s [15]
E	Energy abundance and reliability	Electricity supplied to data centres and labs, TWh/yr, adjusted for unmet interconnection requests [18]
R	Robotics / laboratory automation	Experiments per researcher per year in tracked self-driving-lab domains
B	Biotechnology maturity	Inverse cost per genome and per edited cell line; number of modality classes in human trials
D	Data and measurement capability	Size of openly accessible longitudinal multi-omic cohorts
S	Space accessibility	Inverse launch cost per kg to LEO; orbital research payload mass per year
M	Manufacturing capacity	Advanced-node wafer starts per year; cell- and gene-therapy manufacturing capacity
L	Clinical translation efficiency	Inverse of median time from IND filing to approval

The proxies are proposals. Each must be fixed, together with its data source, before the first forecast is scored.

Why A needs its own sub-index. The METR time horizon is measured mostly on software and machine-learning tasks. An agent that can work autonomously on code for forty hours does not necessarily have equivalent capability in biological or chemical research. Using it alone would turn one domain-specific benchmark into a stand-in for "scientific intelligence" in general. A is therefore defined as a weighted geometric mean of normalized sub-measures:

$$A_t = \prod_k a_{k,t}^{v_k}, \qquad \sum_k v_k = 1,$$

with candidate components: autonomous task horizon (METR); research-level mathematics benchmarks (e.g. FrontierMath); formal theorem proving (share of target statements with machine-checked proofs); scientific hypothesis generation, scored by later experimental confirmation; experimental planning in the wet-lab sciences; coding; and the **cost of a fixed capability level** [26], which measures how cheaply a given level of reasoning can be bought rather than how high the frontier is. The sub-weights v_k are fixed at preregistration, and results are reported both for the composite and for each component. Large formalization efforts of *known* results (for example [28]) are recorded under formal theorem proving, as capability events; they are not discoveries (Section 9).

7.2 Aggregation

The initial gross score is a weighted geometric mean:

$$FTAF_t = A_t^{0.25}\, C_t^{0.15}\, E_t^{0.10}\, R_t^{0.10}\, B_t^{0.15}\, D_t^{0.05}\, S_t^{0.05}\, M_t^{0.05}\, L_t^{0.10}$$

The weights sum to one and are provisional. They reflect the relative importance judged by the author, not an estimate, and must be sensitivity-tested. Because the thesis of this paper is that clinical translation is the binding constraint in biomedicine, a biomedical variant of the index should test a larger weight on L.

A weighted geometric mean is a Cobb–Douglas aggregate with an elasticity of substitution of exactly 1. That means a proportional gain in one component fully offsets the same proportional loss in another, weighted by the exponents; the index only collapses when a component goes to zero. It therefore encodes only weak complementarity. To represent genuine bottlenecks, the framework uses the constant-elasticity-of-substitution (CES) generalization

$$FTAF_t = \left(\sum_j w_j\, X_{j,t}^{\rho} \right)^{1/\rho}, \qquad \sigma = \frac{1}{1-\rho},$$

where $\sigma < 1$ means the components are complements and the weakest subsystem increasingly limits the whole, as in Baumol-type models [9]. The geometric mean is the limit $\rho \to 0$ ($\sigma = 1$), and a strict weakest-link (Leontief) aggregate is the limit $\rho \to -\infty$. The value of σ is a key uncertain parameter and must be varied in the sensitivity analysis.

7.3 Levels, growth rates and acceleration

$FTAF_t$ is a **level** index and equals 1 in 2026 by construction. Evidence for the Saka Law concerns its growth rate, $g_F(t) = d \ln FTAF_t / dt$, and whether that growth rate is rising, $dg_F/dt > 0$. Throughout the paper, "acceleration" refers to the second quantity.

7.4 Separating inputs from outputs

The full index mixes two kinds of component. Some measure **resources committed** to the innovation system; others measure **what the system can do**:

Type	Components
Inputs I	C (compute), E (energy), D (data), M (manufacturing capacity)
Outputs / capabilities O	A (AI capability), R (experiments per researcher), B (biotechnology), S (space access), L (clinical translation)

With a geometric mean, the growth rate of the full index is simply the weighted average of component growth rates, $g_F = \sum_j w_j \, g_j$. An investment boom—more data centres, more power, more fabs—can therefore make $dg_F/dt > 0$ without any change in how efficiently those resources produce new technology. That would be acceleration of *spending*, not of *discovery*, and it is not what the Saka Law claims.

The framework therefore defines two sub-indices with the same weights, renormalized within each group:

$$I_t = C_t^{0.15/0.35} \, E_t^{0.10/0.35} \, D_t^{0.05/0.35} \, M_t^{0.05/0.35},$$

$$O_t = A_t^{0.25/0.65} \, R_t^{0.10/0.65} \, B_t^{0.15/0.65} \, S_t^{0.05/0.65} \, L_t^{0.10/0.65},$$

and an efficiency index analogous to total factor productivity:

$$\Pi_t = \frac{O_t}{I_t^{\eta}}, \qquad g_{\Pi}(t) = g_O(t) - \eta \, g_I(t),$$

where η is the historical elasticity of capabilities with respect to inputs, estimated by regressing $\ln O$ on $\ln I$ over the pre-freeze data and fixed at preregistration. Because that history is short for several components, η is uncertain; results are reported for $\hat{\eta}$ and for $\eta = 1$, and η is sampled in the Monte Carlo.

The full $FTAF_t$ remains a useful descriptive level index. **The test of the hypothesis (prediction P2) is made on O_t and Π_t**, not on $FTAF_t$.

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8. Friction, evidence and accessibility

Three correction terms prevent the framework from confusing technical demonstrations with societal impact. They act at different levels and are therefore kept separate.

8.1 Bottleneck & Risk Index (BPI) — system level

$BPI_t \in [0, 1]$ represents constraints that are **not already measured** by the nine components, such as:

- regulatory delays not captured in L;
- poor scientific reproducibility;

- safety limitations;
- capital constraints;
- geopolitical disruption, including concentration of critical-material supply (for example gallium for GaN power devices, whose exports have been restricted);
- diminishing returns to scaling.

Energy, fabrication and translation limits are already represented by E, M and L and must not be counted again in BPI.

The system-level effective index is

$$F_t^{sys} = FTAF_t\,(1 - BPI_t).$$

8.2 Evidence Maturity Score (EMS) — intervention level

A preclinical result should not carry the same forecasting weight as a replicated human endpoint. EMS is a property of a specific intervention *for a specific indication* (Section 10), not of the technological system as a whole, so it is **not** multiplied into F_t^{sys}. It is used instead to weight evidence when updating the arrival hazards of Section 12.

A provisional ordinal scale is:

hypothesis $<$ in vitro $<$ animal $<$ human observational $<$ controlled trial $<$ large RCT $<$ independent repl

8.3 Diffusion & Accessibility Index (DAI) — intervention level

A therapy that exists but is unaffordable or capacity-constrained has limited population impact. DAI is also intervention-specific and includes:

- price;
- annual treatment capacity;
- geographic diffusion;
- reimbursement / coverage;
- manufacturing scale;
- time to access.

DAI converts the arrival of a therapy into its population-level effect (Section 12); it does not enter the system index.

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9. Scientific throughput rather than paper count

A core prediction of the Saka Law is that the scientific cycle itself should compress.

For a closed experimental loop

$$\text{hypothesis} \to \text{experiment} \to \text{data} \to \text{analysis} \to \text{new hypothesis},$$

define Scientific Cycle Time as the latency of one full loop:

$$SCT = t_{\text{new hypothesis}} - t_{\text{previous hypothesis}},$$

and the Scientific Acceleration Factor

$$SAF_{science} = \frac{SCT_{baseline}}{SCT_t}.$$

Latency alone can be misleading: a laboratory can shorten each loop while running fewer loops. The framework therefore also tracks throughput, Q_t = validated and independently replicated discoveries per unit time in a fixed domain. The primary output is Q_t, not publication volume.

An AI system producing one million low-quality hypotheses is not scientific acceleration if experimental validation and replication remain unchanged.

AI attribution scale. To count "AI-originated" discoveries (prediction P6), each discovery is graded by the role of AI systems in its intellectual work:

Level	Role of AI
AI_0	Auxiliary tool (search, writing, routine code)
AI_1	Data analysis or modelling on a human-designed study
AI_2	Partial hypothesis generation or experimental design, with humans leading
AI_3	Principal scientific contribution: AI proposed the hypothesis *and* designed the decisive experiment or proof; humans executed, checked or supervised
AI_4	Essentially autonomous discovery: AI proposed, executed (in silico or through automated laboratories) and interpreted the work, with humans limited to approval and verification

Levels are assigned from author-contribution statements (e.g. CRediT roles) and methods sections by two independent coders blind to the prediction, with inter-rater agreement (Cohen's κ) reported. Disagreements are resolved to the lower level. Only AI_3 and AI_4 discoveries count toward P6.

Coding rules. Worked calibration cases (repository folder `ai-attribution/`) led to the following rules:

1. *Unit of analysis.* The scale codes discoveries documented in a publication or preprint. Statements in interviews, blogs or social media about how a group uses AI are recorded as **claims** and are not coded, however specific they sound.
2. *Search space.* If AI generates or ranks candidates within a target space chosen by humans, the maximum level is AI_2. AI_3 requires that AI proposed the hypothesis or target itself, not only the best item within a human-defined search.
3. *Speed claims.* Statements about how long a piece of work "would have taken" are not evidence. Acceleration enters the framework only as SCT or Q_t measured from dated records.
4. *Source quality.* Each coded discovery records whether its sources are primary (paper, preprint, registry, formal proof) or secondary (press, interview, social media). Only primary sources can support a level above AI_1.
5. *Formalization is not discovery.* Formalizing a known result is a capability event for the formal-theorem-proving component of A, not a discovery.

6. *Replication of formal proofs.* A machine-checked proof counts as replication for P6 only after humans have confirmed that the formal statement matches the claimed theorem.
7. *Conflicts of interest.* If an AI system used as a coder comes from the same developer as the system being coded, the case is flagged and the human coding is done first.

Domain and importance. Each discovery is assigned to one of three domain groups—(a) mathematics and formal sciences, (b) computational sciences, (c) experimental sciences—because AI can work without physical validation only in (a) and, potentially, in (b); a pooled count would be dominated by mathematics. Each discovery also receives an **importance grade** by the same two-coder procedure: 1 = incremental improvement of a known bound or method; 2 = resolution of a recognized open problem; 3 = resolution of a central problem of the field. P6 is reported as a raw count and as an importance-weighted count, by domain group.

Baseline events. Candidate AI_3/AI_4 discoveries that occur before the preregistration is frozen are logged as baseline events. Several have occurred in mathematics in the months before this version (Section 1), which raises the baseline against which P6 must "at least double"; the baseline period ends on the preregistration date.

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10. Longevity Translation Index (LTI)

Longevity forecasting is particularly vulnerable to overinterpreting animal or biomarker studies.

The LTI therefore tracks interventions through

$$L_0 \to L_1 \to L_2 \to L_3 \to L_4 \to L_5 \to L_6 \to L_7 \to L_8$$

where L_0 = hypothesis, L_1 = in vitro, L_2 = animal, L_3 = large animal, L_4 = Phase I, L_5 = Phase II, L_6 = Phase III, L_7 = approval, and L_8 = clinically meaningful or mortality benefit.

Unit: intervention × indication. An LTI level belongs to the pair *(intervention, indication)*, not to the intervention alone. The same drug can be at very different levels for different questions. Semaglutide, for example, has demonstrated clinical benefit (L_8) in obesity and cardiovascular risk, while the evidence that it slows *ageing* is at L_2–L_3: a lifespan effect in mice confounded by weight loss, and effects on epigenetic clocks in one special population that were not seen in healthy older adults. EMS and the readiness factor G_j (Section 12) are likewise indexed by the pair.

The model should estimate transition probabilities and transition times between levels, using historical baselines such as [21].

Two methodological requirements apply to these estimates and to predictions P7 and P8:

- **Composition bias.** A change in the mix of programmes can move averages without any change in the process: if many simple drugs and few oncology drugs enter development, median time to approval falls even if nothing improved. Comparisons are therefore made *within* strata of the same therapeutic area and the same modality (small molecule, biologic, cell therapy, gene therapy, etc.), between cohorts defined by the year of IND filing, and then combined with fixed stratum weights. **Repurposed drugs**—already approved for another indication—are analysed as a separate stratum, because they skip early phases and would otherwise make translation look faster by construction.
- **Censoring.** Many programmes in recent cohorts will still be unresolved when a window closes. Excluding them would bias recent times downward (only fast successes are

observed). Times to approval are therefore estimated with survival analysis—Kaplan–Meier curves and Cox models with therapeutic area and modality as covariates—and phase progression with a multistate model ($L_4 \to L_5 \to L_6 \to L_7$, with failure as a competing absorbing state).

A major acceleration in ageing-biology papers with no reduction in $L_1 \to L_8$ translation time, and no improvement in transition probabilities, would argue against strong biomedical recursive acceleration.

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11. Longevity Escape Velocity proxy

Longevity escape velocity (LEV) [23] concerns whether a person's remaining healthy life stops shrinking as they age. Version 0.3.1 defined the period proxy as the raw time derivative of healthy-life expectancy at a fixed age, with threshold 1. That threshold is not the escape condition, and this version replaces it.

Why the raw derivative with threshold 1 is wrong. Let $HALE_x(t)$ be the period healthy-life expectancy at age x in calendar year t, computed from that year's age-specific health and mortality rates. Consider a person whose risk matches the population's at each age. Their remaining healthy life is $h(t) = HALE_{a(t)}(t)$, with $da/dt = 1$, so

$$\frac{dh}{dt} = \frac{\partial\, HALE_x}{\partial t} + \frac{\partial\, HALE_x}{\partial x}.$$

Escape velocity is $dh/dt \geq 0$, which requires

$$\frac{\partial\, HALE_x}{\partial t} \;\geq\; -\frac{\partial\, HALE_x}{\partial x}.$$

The right-hand side—how much remaining healthy life is lost per year of age in a single period table—is usually **less than 1** at older ages, because surviving a year is itself selective. A threshold of 1 on the time derivative alone is therefore stricter than the escape condition, and its interpretation as "one year gained per year lived" mixes the period and individual views.

Normalized period definition (used by this framework). Define

$$LEV_x(t) = \frac{\partial_t\, HALE_x(t)}{-\,\partial_x\, HALE_x(t)}.$$

Interpretation:

- $LEV_x < 0$: healthy-life expectancy at age x is falling over calendar time;
- $0 < LEV_x < 1$: medical progress offsets a fraction LEV_x of the healthy life that a person of age x loses by ageing one year;
- $LEV_x \geq 1$: medical progress fully offsets that loss—the operational definition of longevity escape velocity in this framework.

With this normalization the period proxy and the individual condition $dh/dt \geq 0$ coincide for an individual with population-average risk, so the threshold is the same in both views. They still differ for any individual whose risk departs from the population's, and the period proxy assumes that current age-specific rates describe the future. The numerator $\partial_t HALE_x$ is also reported on its own, as the raw speed of the healthy-life frontier.

Estimation. The denominator is estimated by finite differences across adjacent ages in the same period table. Most HALE series are published in five-year age groups, so single-year values must be interpolated; the interpolation method is fixed at preregistration. Both derivatives are noisy, and LEV_x is reported with an interval, not as a point.

Reference point, not an estimate. Record period life expectancy *at birth* has risen by about 0.25 years per year for more than 160 years [22]. That is a reference for the frontier of total life expectancy; it is not an estimate of LEV_{65}, which is a different series (healthy rather than total life, at age 65 rather than at birth, normalized by the age gradient, and in a given population rather than the record-holding one). Healthy-life expectancy has generally grown more slowly than total life expectancy. The current empirical value of the proxy must be estimated from fixed-age HALE series, and establishing that baseline is one of the first data tasks of the framework.

This is an operational forecasting proxy rather than an accepted clinical metric. The framework does not claim that $LEV_x \geq 1$ has been achieved.

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12. The Technology Survival Ladder

Long-range longevity forecasts should not require a single future "cure for ageing".

Instead, an individual may encounter sequential generations of treatment:

$$\text{Therapy}_1 \rightarrow \Delta HLE_1 \rightarrow \text{Therapy}_2 \rightarrow \Delta HLE_2 \rightarrow \text{Therapy}_3$$

If earlier interventions preserve enough function to reach later, more capable interventions, a ladder effect emerges.

For intervention class j—understood as a pair (intervention class, indication), as in Section 10—define the arrival hazard of clinical availability:

$$\lambda_j(t) = \lambda_{0,j} \, [F_t^{sys}]^{\alpha_j} \, G_j(t),$$

where $\lambda_{0,j}$ is the baseline hazard in 2026, α_j the sensitivity of class j to system-wide technological progress, and $G_j(t)$ a readiness factor determined by the class's current LTI level and the historical transition probabilities out of that level. Evidence about the class, weighted by EMS, updates G_j.

The cumulative arrival probability from 2026 ($t = 0$) to time T is

$$P_j(T) = 1 - \exp\left(- \int_0^T \lambda_j(t) \, dt \right).$$

Population impact is then $P_j(T)$ multiplied by the class's projected DAI_j.

In version 0.3.4, $\lambda_{0,j}$, α_j and G_j are not estimated. Doing so requires the LTI transition data described in Section 10. This should be interpreted as technology-arrival forecasting, not as a personalized survival probability.

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13. Space as an experimental multiplier

Space is included with a relatively modest weight because biomedical progress does not require orbital laboratories. However, cheaper and more frequent launch could add experimental regimes difficult to reproduce on Earth:

- prolonged microgravity;
- altered fluid dynamics;
- radiation environments;
- protein crystallization;
- tissue and organoid growth;
- closed-loop autonomous biological experiments.

If reusable launch systems substantially reduce cost and increase cadence, the space term S_t may undergo a measurable change point.

The value of space in the framework is therefore not speculative colonization; it is **expansion of experimental state space**. With a weight of 0.05, however, its effect on the index is small, and it could be moved to future work without changing the main results.

---

14. Energy as a cognitive-infrastructure constraint

AI systems transform electricity into computational work. Large-scale scientific agents, simulation and automated laboratories therefore depend on physical energy infrastructure.

The IEA's 2026 update emphasizes both rapidly improving energy efficiency per AI task and rapidly increasing aggregate demand from more intensive applications [18].

This creates a feedback structure:

$$\text{Energy} \rightarrow \text{Compute} \rightarrow \text{AI} \rightarrow \text{Science} \rightarrow \text{better energy technology}$$

Such a loop can accelerate, but transmission grids, generation capacity, transformers, storage and permitting can impose long delays.

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15. Twenty-year scenario: 2026–2046

This section is explicitly speculative.

A naive continuation of recent AI growth rates would produce implausibly enormous multipliers. The Saka framework therefore assumes declining log-growth rates and explicit bottlenecks.

A reasonable scenario family for FTAF should include:

- **conservative:** rapid near-term growth followed by strong saturation;
- **central:** sustained AI/science acceleration with progressively declining growth rates;
- **aggressive:** major breakthroughs in AI, automation, energy or manufacturing that extend the high-growth regime.

Version 0.3.4 does not attach numbers to these scenarios. They will be quantified once the proxies of Section 7.1 have been populated.

The central qualitative expectation for 2046 is not "immortality". It is increased probability of:

- highly longitudinal personalized medicine;
- earlier multi-modal cancer detection;
- broader use of in-vivo gene editing;
- organoids as therapeutic decision tools in selected cancers;
- clinically useful regeneration of multiple tissues;
- substantially increased organ replacement options;
- autonomous laboratories as standard research infrastructure;
- at least some human interventions that restore age-related function in specific tissues.

The most important uncertainty is whether these capabilities compress **clinical translation time**, not merely discovery time.

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16. 2046–2076 and the longevity question

Forecast uncertainty becomes extreme beyond 2046.

The relevant comparison is between the rate of ageing and the rate of medical progress, summarized by the proxy $LEV_x(t)$ of Section 11.

By 2076, a successful recursive-acceleration regime could plausibly produce periodic repair across multiple biological subsystems rather than a single rejuvenation treatment.

However, several problems may remain especially difficult:

- preserving brain information while rejuvenating neural tissue;
- preventing cancer under increased regenerative capacity;
- maintaining long-term genomic and epigenomic stability;
- controlling immune ageing without creating autoimmunity;
- proving that biomarker changes translate into morbidity and mortality reduction.

The framework therefore treats “systemic rejuvenation” and “longevity escape velocity” as high-uncertainty events rather than inevitable outcomes.

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17. Predictions that make the framework falsifiable

The qualitative predictions of version 0.1 are replaced by dated, quantitative ones.

This version is not the preregistration. Version 0.3.4 fixes the *form* of each prediction and the rule for deriving its threshold, but not the thresholds themselves. The preregistration is made when that rule has been applied to historical data and the resulting numbers are frozen, with a timestamp, in a public file (planned: `forecasts/preregistration-2026.json`), before any data from the tested windows are examined.

Windows. Each window starts on the date the preregistration is frozen, not on 1 January 2026; data observed before that date are used only to estimate baselines. The window labels below (2026–2031, 2026–2036) are nominal and assume a freeze in late 2026.

Status of the thresholds. The numbers below (for example $TAR > 1.5$ or a 50% fall in SCT) are **placeholders chosen as plausible magnitudes, not derived values**. Before preregistration, each threshold must be derived from two inputs:

1. *Historical variability*: the distribution of the same statistic over past windows of the same length, so that a threshold is exceeded with at most 5% probability if the historical regime continues; and
2. *Minimum relevant effect*: the smallest change that would matter scientifically or clinically.

The preregistered threshold is the larger of the two. This turns the predictions from subjective cut-offs into statistical tests.

Multiplicity. Several predictions succeed if a criterion holds "in at least k" of many domains or strata (P1, P3, P5, P7, P8). With enough strata, some will cross any per-stratum threshold by chance. The 5% rule above is therefore applied to the **whole statement**, not to each stratum: the threshold is set so that the event "at least k of the m tested domains/strata pass" has at most 5% probability under the historical regime, obtained by simulation from the historical variability or, where simulation is not feasible, by a Holm correction across strata. The list of domains and strata, and hence m, is fixed at preregistration; adding strata afterwards is not allowed. All eight predictions are reported whatever their outcome, and no single success is presented as confirming the framework.

Power. Before the thresholds are frozen, each prediction is simulated under two regimes: the historical regime continuing, and a minimally relevant acceleration. The resulting statistical power is published with the preregistration. This matters most for P2 and P4, which estimate a change in a growth rate—a second derivative of a logarithm—from a short window of noisy, autocorrelated data. A prediction with low power (below 50%) is either given a longer window or marked as exploratory, so that a null result is not misread as evidence against the hypothesis.

Intervals. All estimates are reported with 50%, 80%, 90% and 95% intervals. The **90% interval** is preregistered as the decision criterion: "lower bound above 1" or "interval excluding 0" below refers to it.

Supporting predictions. If recursive acceleration is real, then:

#	Prediction	Window	Supports the hypothesis if…
P1	Cross-domain acceleration	2026–2031	$TAR_i > 1.5$ (domain-level, Section 6), with the lower bound of its 90% interval above 1, in at least 3 of the HTAB domains
P2	Rising growth rate of capabilities and efficiency	2026–2031	Estimated $dg_O/dt > 0$ **and** $dg_\Pi/dt > 0$ (Section 7.4), each with its 90% interval excluding 0; a rise in dg_F/dt driven only by inputs does not count
P3	Research productivity reverses	2026–2036	Research productivity (in the sense of [6]), measured against R^{total} (Section 4), rises in at least 2 of the domains studied there

#	Prediction	Window	Supports the hypothesis if…
P4	Autonomy accelerates	2026–2031	The METR 50% time-horizon doubling time over the window is **shorter** than over the pre-freeze baseline period, with the 90% interval of their ratio below 1; and the non-software components of the composite A index keep rising
P5	Faster experimental loops	2026–2031	Median SCT falls by at least 50% in at least 2 tracked self-driving-lab domains, with no fall in Q_t
P6	Replicated AI discoveries	2026–2031	The annual number of independently replicated discoveries graded AI_3 or AI_4 (Section 9), relative to the baseline frozen at preregistration, at least doubles **and** exceeds a preregistered absolute minimum, in both the raw and the importance-weighted count; **and** the count in the experimental sciences rises. A rise confined to mathematics is reported as partial support only

#	Prediction	Window	Supports the hypothesis if…
P7	Faster translation	2026–2036	Median IND-to-approval time, estimated by survival analysis within therapeutic area × modality strata, excluding repurposed drugs (Section 10), falls by at least 20% relative to 2015–2025 IND cohorts in at least one therapeutic area
P8	Better clinical success	2026–2036	Phase I-to-approval success probability, estimated with a multistate model within therapeutic area × modality strata, excluding repurposed drugs (Section 10), improves by at least 30% relative to [21] in at least one therapeutic area

Exploratory statistic (not a prediction). For the cost of a fixed AI capability level [26], the rate of decline is fastest when a level first becomes reachable and slows afterwards; that within-level slowdown is an ordinary learning curve. The statistic relevant to the hypothesis is the **debut rate across cohorts**: whether the initial rate of cost decline rises for successive performance levels as they appear. It is reported alongside P2 and P4 but, because the series is short, it is marked exploratory and not counted in the multiplicity correction.

Refuting outcomes. The hypothesis should be weakened or rejected as a useful forecasting framework if, by the end of the relevant window:

- TAR_i is not distinguishable from 1 in a majority of HTAB domains (P1 fails);
- AI benchmark and autonomy metrics keep improving while SCT, Q_t and replication rates do not (P5 and P6 fail);
- clinical translation times and success probabilities are statistically unchanged (P7 and P8 fail);
- $LEV_{65}(t)$ for the largest high-income populations shows no increase over its own 2000–2025 trend through 2046;
- persistent energy, fabrication, regulatory or biological bottlenecks explain most of the shortfall.

Failure of P7, P8 and the LEV criterion while P1–P6 succeed would support recursive acceleration in computation and discovery but refute its extension to biomedicine—itself an informative result.

18. Forecasting protocol

Future versions should use a preregistered update process.

Quarterly

Update:

- AI capability and autonomy;
- compute stock and price-performance;
- energy availability and data-centre constraints;
- laboratory automation;
- major clinical translation milestones.

Annually

Fit:

- linear;
- exponential;
- power-law;
- logistic;
- piecewise/change-point models.

Evaluate using:

- rolling out-of-sample error;
- AIC/BIC;
- residual diagnostics;
- calibration.

Three technical cautions apply:

- A power law $y = at^b$ depends on where $t = 0$ is placed; the origin must be fixed in advance and reported.
- Logistic curves fitted before the inflection point are poorly identified: the ceiling L is essentially unconstrained. Logistic fits should report the profile likelihood of L rather than a point estimate.
- AIC and BIC can only compare models fitted to the same response on the same scale with the same error structure; models fitted in levels and in logs must be compared by out-of-sample error instead.

Monte Carlo

Simulate at least 10^5 futures with distributions over:

- growth-rate decay;
- breakthrough probabilities;
- the elasticity of substitution σ between subsystems;
- the input elasticity η of the efficiency index (Section 7.4);
- energy constraints;
- AI reliability;
- experimental throughput;
- regulatory timelines;
- clinical success probabilities.

Predictions should be scored retrospectively using Brier scores [24] where possible.

Data sources

Candidate public data sources for each component, for the HTAB baseline and for the LEV proxy are listed in the technical specification (`model-spec.md`, Section 13). Three inputs—Scientific Cycle Time, validated throughput Q_t and the AI_0–AI_4 attribution—have no existing dataset and must be built by the project.

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19. Discussion

The principal contribution of the Saka Law framework is not a claim that technology will grow exponentially forever, nor the idea that knowledge feeds back into its own production, which has a long history in growth economics [3–5, 9]. It is the proposal that **research productivity itself should be measured as an endogenous technological variable, across several physical and institutional subsystems, against a preregistered baseline**.

This changes long-range forecasting.

In a non-recursive model, the future is estimated mainly from the current stock of technology.

In a recursive model:

$$\text{Future technology} = f\big(\text{current technology},\ \text{technology's ability to improve discovery}\big)$$

If scientific cognition, experiment execution, measurement and manufacturing all become increasingly automated, technological progress may depart from historical baselines.

But a recursive process is not necessarily an explosive one. Real systems can settle into logistic, piecewise or bottleneck-limited regimes, and the evidence of falling research productivity [6, 7] means the burden of proof lies with the acceleration hypothesis.

The empirical task is therefore to measure the curvature of $\ln T$, not of T.

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20. Conclusion

The Saka Law is proposed as a testable hypothesis of recursive technological acceleration:

> **When technology materially improves the processes that generate, validate and deploy new technology, the growth rate of technological progress can rise relative to its historical baseline, until constrained by physical, biological, economic or institutional bottlenecks.**

The Future Technology Acceleration Framework attempts to measure this effect across AI, compute, energy, robotics, biotechnology, measurement, space, manufacturing and clinical translation.

Its longevity application replaces deterministic predictions of "curing ageing" with a measurable sequence:

scientific acceleration $\rightarrow$ biomedical translation $\rightarrow$ functional repair $\rightarrow$ healthy-life extension $\rightarrow$ access to

The hypothesis will become scientifically useful only if its metrics are populated with reproducible data, its forecasts are preregistered, and its failures are recorded as rigorously as its successes.

References

1. OpenAI. **On the Navier–Stokes Millennium Prize Problem.** 8 September 2026. https://openai.com/index/navier-stokes-solution/
2. Quanta Magazine. **AI Has Solved One of Math's $1 Million Millennium Prize Problems.** 8 September 2026. https://www.quantamagazine.org/ai-has-solved-one-of-maths-1-million-millennium-prize-problems-20260908/
3. Romer, P. M. **Endogenous technological change.** *Journal of Political Economy* 98(5), S71–S102 (1990).
4. Jones, C. I. **R&D-based models of economic growth.** *Journal of Political Economy* 103(4), 759–784 (1995).
5. Kremer, M. **Population growth and technological change: One million B.C. to 1990.** *Quarterly Journal of Economics* 108(3), 681–716 (1993).
6. Bloom, N., Jones, C. I., Van Reenen, J. & Webb, M. **Are ideas getting harder to find?** *American Economic Review* 110(4), 1104–1144 (2020).
7. Scannell, J. W., Blanckley, A., Boldon, H. & Warrington, B. **Diagnosing the decline in pharmaceutical R&D efficiency.** *Nature Reviews Drug Discovery* 11, 191–200 (2012).
8. Kurzweil, R. **The Law of Accelerating Returns.** Essay, 2001.
9. Aghion, P., Jones, B. F. & Jones, C. I. **Artificial intelligence and economic growth.** In Agrawal, A., Gans, J. & Goldfarb, A. (eds.), *The Economics of Artificial Intelligence: An Agenda*, University of Chicago Press (2019).
10. Nordhaus, W. D. **Are we approaching an economic singularity? Information technology and the future of economic growth.** *American Economic Journal: Macroeconomics* 13(1), 299–332 (2021).
11. Davidson, T. **Could advanced AI drive explosive economic growth?** Open Philanthropy report (2021).
12. Erdil, E. & Besiroglu, T. **Explosive growth from AI automation: A review of the arguments.** arXiv:2309.11690 (2023).
13. Nagy, B., Farmer, J. D., Bui, Q. M. & Trancik, J. E. **Statistical basis for predicting technological progress.** *PLoS ONE* 8(2), e52669 (2013).
14. Farmer, J. D. & Lafond, F. **How predictable is technological progress?** *Research Policy* 45(3), 647–665 (2016).
15. Epoch AI. **Trends in Artificial Intelligence.** Updated 2026. https://epoch.ai/trends
16. METR. **Task-Completion Time Horizons of Frontier AI Models.** Updated 8 May 2026. https://metr.org/time-horizons/
17. Canty, R. B. & Abolhasani, M. **The past, present and future of self-driving laboratories.** *Nature Reviews Chemistry* 10, 523–537 (2026). https://doi.org/10.1038/s41570-026-00847-2
18. International Energy Agency. **Key Questions on Energy and AI.** 2026. https://www.iea.org/reports/ke questions-on-energy-and-ai/executive-summary
19. Life Biosciences. **Life Biosciences Announces First Patient Dosed in Phase 1 Trial of ER-100 for Optic Neuropathies.** 9 June 2026. https://www.lifebiosciences.com/life-biosciences-announces-first-patient-dosed-in-phase-1-trial-of-er-100-for-optic-neuropathies/

20. ClinicalTrials.gov. **NCT07290244: Evaluating ER-100 for Safety in People With Glaucoma or Non-Arteritic Anterior Ischemic Optic Neuropathy.** https://clinicaltrials.gov/study/NCT07290244

21. Wong, C. H., Siah, K. W. & Lo, A. W. **Estimation of clinical trial success rates and related parameters.** *Biostatistics* 20(2), 273–286 (2019).

22. Oeppen, J. & Vaupel, J. W. **Broken limits to life expectancy.** *Science* 296(5570), 1029–1031 (2002).

23. de Grey, A. D. N. J. **Escape velocity: Why the prospect of extreme human life extension matters now.** *PLoS Biology* 2(6), e187 (2004).

24. Brier, G. W. **Verification of forecasts expressed in terms of probability.** *Monthly Weather Review* 78(1), 1–3 (1950).

25. International Energy Agency. **Energy and AI.** 2025. https://www.iea.org/reports/energy-and-ai

26. Epoch AI. **The plunging price of thought.** 22 September 2026. https://epoch.ai/publications/the-plunging-price-of-thought

27. Anthropic. **Claude has improved on a longstanding lower bound for the fraction of zeros of the Riemann zeta function that satisfy the Riemann hypothesis.** 10 August 2026. https://www.anthropic.com/research/riemann-zeta

28. Anthropic. **Formalizing Fermat's Last Theorem.** 4 September 2026. https://www.anthropic.com/res fermats-last-theorem

29. Wright, T. P. **Factors affecting the cost of airplanes.** *Journal of the Aeronautical Sciences* 3(4), 122–128 (1936).

Correction in version 0.3.4

- **Section 4, definition of research productivity.** Version 0.3.3 wrote research productivity as $\dot{A}/R^{\lambda} = \delta A^{\phi}$ and called a rising ϕ "equivalent" to rising productivity. The quantity measured in [6] is growth per unit of input, $g_A/R^{\lambda} = \delta A^{\phi-1}$. The condition for recursive acceleration is now $\beta = 1 - \phi < 0$, which is consistent with the super-exponential signature stated earlier in the section and with P3. No prediction changes.

Changes from version 0.3.2

- **LTI, EMS and G_j indexed by intervention × indication** (Sections 8.2, 10, 12), motivated by GLP-1 receptor agonists. Repurposed drugs form a separate stratum in P7 and P8.
- **Maturity-aligned HTAB and TAR** (Sections 5, 6): series record introduction date and cumulative production; TAR_i^{mat} compares a young technology with others at the same age. Motivated by the reported fall in the cost of AI capability [26].
- **Cost of a fixed capability level** added as a sub-measure of A; an exploratory **debut-rate** statistic added to Section 17.
- **AI attribution coding rules** (Section 9): unit of analysis, search-space rule, speed claims, source quality, formalization, replication of formal proofs, conflicts of interest. Developed from calibration cases in the repository.
- **P6** now counts from a baseline frozen at preregistration, by domain group and with an importance grade, and requires a rise in the experimental sciences for full support.

- HTAB candidate series: power electronics and cost of AI capability. BPI: critical-material supply concentration.
- Section 1 notes other AI results in mathematics (August–September 2026).

Changes from version 0.3.1

- **Inputs vs outputs.** Added Section 7.4: the state vector is split into resource inputs (C, E, D, M) and capabilities (A, R, B, S, L), with an efficiency index $\Pi_t = O_t / I_t^{\eta}$. P2 is now tested on O_t and Π_t, so that an investment boom alone cannot satisfy it.
- **P4 now tests acceleration.** The previous criterion (doubling time at or below 12 months) was satisfied by a plain exponential, contradicting Section 4. P4 now requires the METR doubling time to shorten relative to the pre-freeze baseline.
- **LEV proxy normalized.** The raw time derivative of $HALE_x$ with threshold 1 is not the escape condition. LEV_x is now the time derivative divided by the loss of healthy life per year of age, so that $LEV_x \geq 1$ is equivalent to $dh/dt \geq 0$ for an individual with population-average risk.
- **Research input with automation.** Section 4 now defines R^{total} (deflated research expenditure including compute and lab automation) as the primary denominator for research productivity; productivity measured against human researchers alone does not count as evidence. P3 uses R^{total}.
- **Domain-level TAR.** Section 6 defines TAR_i against each domain's own history, used by P1; differences replace ratios for stagnating series.
- **Multiplicity and power.** Thresholds for "at least k of m" predictions control the error rate of the whole statement; the domain and stratum lists are fixed at preregistration; a power analysis under historical and minimally-accelerated regimes is published with the preregistration.
- **P6** requires a preregistered absolute minimum in addition to doubling.
- **Navier–Stokes.** Specified that the 8 September 2026 result concerns the forced equations (a breakdown alternative accepted by the Clay formulation) and that the unforced case remains open.
- Added a pointer to candidate data sources (model specification, Section 13); fixed leftover references to earlier version numbers in Sections 12 and 15.

Changes from version 0.3

- Intervals: 90% added to the reported set (50/80/90/95%), since it is the decision criterion.
- Stated explicitly that this version is not the preregistration; windows start when the thresholds are frozen, not on 1 January 2026.
- Added the AI_0–AI_4 attribution scale, with coding procedure; P6 counts only AI_3 and AI_4.
- P7 and P8: comparisons within therapeutic area × modality strata and IND-year cohorts to control composition bias; survival analysis (Kaplan–Meier, Cox) and multistate models to handle censored programmes.

Changes from version 0.2

- LEV: the 0.25 years/year trend is now presented as a reference for record life expectancy at birth, not as an estimate of LEV_{65}; the fixed-age HALE baseline is to be estimated from data. The two LEV forms are described as related, not equivalent. The LEV refutation criterion is now relative to its own historical trend.
- Prediction thresholds are marked as placeholders, with a rule for deriving them from historical variability and minimum relevant effect.

- Intervals: 50/80/95% reported; the 90% interval is preregistered as the decision criterion (previously 80%).
- A is now a composite index; the METR time horizon is one component rather than the whole proxy.

Changes from version 0.1

- Restated the law as a claim about the **growth rate** of technology (super-exponential growth, rising research productivity) instead of $\ddot{T} > 0$, which any exponential satisfies.
- Added related work (idea-production function, falling research productivity, Eroom's law, AI and explosive growth) and a subsection of evidence against the hypothesis.
- Corrected the LEV proxy: it is now defined on period healthy-life expectancy at a fixed age with threshold 1, and the equivalent individual definition with threshold 0 is stated; added the historical reference of about 0.25 years per year.
- Clarified that a weighted geometric mean has unit elasticity of substitution and added a CES generalization with σ as a sensitivity parameter.
- Separated system-level friction (BPI) from intervention-level evidence (EMS) and accessibility (DAI); removed double counting of energy, fabrication and translation in BPI.
- Defined $G_j(t)$ in the arrival hazard (previously M_j, which clashed with manufacturing).
- Added a table of candidate operational proxies for the nine components.
- Replaced qualitative predictions with dated, quantitative, preregistrable thresholds.
- Added selection-bias, uncertainty-interval and model-comparison cautions to HTAB, TAR and the fitting protocol.
- Made the Navier–Stokes and ER-100 descriptions precise.
- Fixed mathematical notation so equations render on GitHub.

Disclosure

This document is a conceptual forecasting framework developed through human–AI collaboration. Named indices, weights, thresholds and long-range scenarios are proposals for testing, not established scientific standards. Biomedical sections are not medical advice and do not assert that ageing has been or will be cured.

Technical appendix: Model Specification v0.3.4

1. State vector

At time t, define

$$X_t = [A, C, E, R, B, D, S, M, L]$$

where:

Symbol	Component	Candidate operational proxy
A	AI / scientific-intelligence capability	Composite index (Section 1.1); METR 50%-success task horizon is one component
C	Compute availability	Installed AI compute stock, FLOP/s
E	Energy abundance	Electricity supplied to data centres and labs, TWh/yr, adjusted for unmet interconnection requests
R	Robotics and laboratory automation	Experiments per researcher per year in tracked self-driving-lab domains
B	Biotechnology maturity	Inverse cost per genome and per edited cell line; modality classes in human trials
D	Data and measurement capability	Size of openly accessible longitudinal multi-omic cohorts
S	Space accessibility / orbital experimental capacity	Inverse launch cost per kg to LEO; orbital research payload mass per year
M	Manufacturing capability	Advanced-node wafer starts per year; cell- and gene-therapy manufacturing capacity
L	Clinical translation efficiency	Inverse of median time from IND filing to approval

Each component is normalized to a baseline year, initially 2026 = 1. Proxies and data sources must be fixed before the first forecast is scored.

1.1 Composite scientific-capability index METR time horizons are measured mostly on software/ML tasks and must not stand in for scientific capability in general. Define

$$A_t = \prod_k a_{k,t}^{v_k}, \qquad \sum_k v_k = 1,$$

over normalized sub-measures a_k: autonomous task horizon (METR); research-level mathematics (e.g. FrontierMath); formal theorem proving; scientific hypothesis generation scored by later experimental confirmation; wet-lab experimental planning; coding; cost of a fixed capability level (Epoch AI 2026). Fix v_k at preregistration; report the composite and each

component. Formalizations of known results are capability events for formal theorem proving, not discoveries.

2. Gross acceleration index

2.1 Default: weighted geometric mean

$$FTAF_t = A_t^{0.25}\, C_t^{0.15}\, E_t^{0.10}\, R_t^{0.10}\, B_t^{0.15}\, D_t^{0.05}\, S_t^{0.05}\, M_t^{0.05}\, L_t^{0.10}$$

Weights sum to 1, are provisional and must be sensitivity-tested (including a biomedical variant with a larger weight on L).

A weighted geometric mean is a Cobb–Douglas aggregate with elasticity of substitution $\sigma = 1$: a proportional gain in one component fully offsets a weighted proportional loss in another. It only collapses when a component reaches zero, so it encodes weak complementarity only.

2.2 Bottleneck-sensitive: CES

$$FTAF_t = \left(\sum_j w_j\, X_{j,t}^{\rho} \right)^{1/\rho}, \qquad \sigma = \frac{1}{1-\rho}$$

- $\rho \to 0$ ($\sigma = 1$): geometric mean (2.1).
- $\rho < 0$ ($\sigma < 1$): complements; the weakest subsystem increasingly limits the index.
- $\rho \to -\infty$: Leontief weakest-link, $FTAF_t = \min_j X_{j,t}$.

σ is a key uncertain parameter and is sampled in the Monte Carlo (Section 11).

2.3 Level, growth rate, acceleration $FTAF_t$ is a level and equals 1 in 2026 by construction. Define its growth rate

$$g_F(t) = \frac{d \ln FTAF_t}{dt}.$$

"Acceleration" in this framework means $dg_F/dt > 0$ (super-exponential growth), not $d^2FTAF/dt^2 > 0$, which any exponential satisfies.

2.4 Inputs, outputs and efficiency Split the state vector into resource inputs and capabilities:

- Inputs I: C, E, D, M.
- Outputs / capabilities O: A, R, B, S, L.

With the default weights renormalized within each group:

$$I_t = C_t^{0.15/0.35}\, E_t^{0.10/0.35}\, D_t^{0.05/0.35}\, M_t^{0.05/0.35},$$

$$O_t = A_t^{0.25/0.65}\, R_t^{0.10/0.65}\, B_t^{0.15/0.65}\, S_t^{0.05/0.65}\, L_t^{0.10/0.65}$$

Efficiency index (TFP analogue):

$$\Pi_t = \frac{O_t}{I_t^{\eta}}, \qquad g_\Pi = g_O - \eta\, g_I$$

- η: historical elasticity of O with respect to I, from a regression of $\ln O$ on $\ln I$ over pre-freeze data; fixed at preregistration; results also reported for $\eta = 1$; sampled in the Monte Carlo.
- Because $g_F = \sum_j w_j g_j$ for the geometric mean, $dg_F/dt > 0$ can be produced by input growth alone. **P2 is tested on O_t and Π_t, not on $FTAF_t$.**

3. Friction, evidence and diffusion

The three correction terms act at different levels.

System level. $BPI_t \in [0, 1]$ is the Bottleneck & Risk Index. It covers only constraints not already measured by X_t (e.g. reproducibility, safety, capital, geopolitics including critical-material supply concentration such as gallium, regulation beyond L). Energy, fabrication and translation limits are in E, M and L and must not be counted again.

$$F_t^{sys} = FTAF_t \, (1 - BPI_t)$$

Intervention level. For intervention or claim j:

- $EMS_j \in [0, 1]$, the Evidence Maturity Score (Section 8), weights evidence when updating the readiness factor G_j of the arrival hazard (Section 10).
- $DAI_j \in [0, 1]$, the Diffusion & Accessibility Index, converts arrival probability into population impact.

EMS and DAI are not multiplied into F_t^{sys}.

4. Historical Technology Acceleration Baseline

For historical metric i, estimate a log-growth rate. The two-point form is

$$k_i = \frac{\ln\left(V_{i,t_2}/V_{i,t_1}\right)}{t_2 - t_1},$$

but k_i should in practice be estimated by regression of $\ln V_i$ on t over the full series, with a standard error. For decreasing-cost metrics, invert the ratio.

The historical baseline is

$$k_{HTAB} = \sum_i w_i \, k_i,$$

and the Technology Acceleration Ratio is defined per domain, against that domain's own history,

$$TAR_{i,t} = \frac{k_{i,current,t}}{k_{i,historical}},$$

and in aggregate,

$$TAR_t = \frac{\sum_i w_i \, k_{i,current,t}}{k_{HTAB}}.$$

Domain-level predictions (P1) use TAR_i. For series whose historical rate is near zero or negative, report the difference $k_{i,current} - k_{i,historical}$ instead of the ratio.

Maturity alignment. Record each series' introduction date and, where available, cumulative production. For a young technology, also report

$$TAR_i^{mat}(t) = \frac{k_{i,current}(t)}{\bar{k}_{HTAB}(a_i)},$$

with a_i the technology's age and $\bar{k}_{HTAB}(a)$ the weighted HTAB log-growth rate at the same age (or the same multiples of cumulative production; Wright 1936). Acceleration of a young technology requires $TAR_i^{mat} > 1$, not only a high calendar-aligned TAR_i.

Interpretation:

- $TAR < 1$: slower than historical baseline
- $TAR \approx 1$: similar to historical baseline
- $TAR > 1$: accelerated regime

Requirements:

- The series list must include domains that stagnated (e.g. drugs approved per R&D dollar, crop yields, transport speed, construction productivity), not only known success stories.
- Series and weights w_i are fixed before current data are compared.
- Candidate series include power-electronics cost per watt and efficiency (Si, SiC, GaN) and the cost of a fixed AI capability level.
- TAR is always reported with an uncertainty interval; acceleration requires its lower bound to exceed 1.

4.1 Research productivity with automated inputs Research productivity (Bloom et al. 2020) is output growth per unit of effective research input. When AI substitutes for researchers, define:

- R^{total} (**primary**): deflated research expenditure, including compute, cloud, model access and laboratory automation bought for research.
- R^{human} (secondary): research personnel only. Reported to show substitution; a productivity rise measured only against R^{human} is **not** evidence for the hypothesis.

Deflator and expenditure classification are fixed at preregistration. P3 uses R^{total}.

5. Growth-model comparison

For every major time series, fit at least:

1. Linear:
$$y(t) = a + bt$$
2. Exponential:
$$y(t) = ae^{kt}$$
3. Power law:
$$y(t) = at^b$$
4. Logistic:
$$y(t) = \frac{L}{1 + e^{-k(t-t_0)}}$$

5. Piecewise / change-point models.

Model selection should use rolling out-of-sample error plus AIC/BIC and residual diagnostics. No permanent exponential assumption is allowed.

Cautions:

- The power law depends on the origin of t; fix it in advance and report it.
- Logistic fits before the inflection point leave the ceiling L essentially unidentified; report its profile likelihood, not a point estimate.
- AIC/BIC are only comparable between models fitted to the same response, on the same scale, with the same error structure. Models fitted in levels and in logs are compared by out-of-sample error only.

6. Scientific translation

For a closed experimental loop, define Scientific Cycle Time (latency of one loop):

$$SCT = t_{\text{new hypothesis}} - t_{\text{prior hypothesis}}$$

and the Scientific Acceleration Factor

$$SAF_{science} = \frac{SCT_{baseline}}{SCT_t}.$$

Because latency can fall while the number of loops falls too, also track throughput:

Q_t = validated and independently replicated discoveries per unit time in a fixed domain.

Define the Translation Acceleration Factor

$$TAF = \frac{T_{baseline,\ \text{discovery}\to\text{clinic}}}{T_{current,\ \text{discovery}\to\text{clinic}}}.$$

These variables attempt to measure conversion of computational progress into physical and clinical progress.

6.1 AI attribution scale Grade each discovery by the role of AI in its intellectual work:

- AI_0: auxiliary tool (search, writing, routine code);
- AI_1: data analysis or modelling on a human-designed study;
- AI_2: partial hypothesis generation or experimental design, humans leading;
- AI_3: principal contribution — AI proposed the hypothesis and designed the decisive experiment or proof;
- AI_4: essentially autonomous — AI proposed, executed (in silico or via automated labs) and interpreted; humans approve and verify.

Assigned from CRediT statements and methods sections by two independent coders blind to the prediction; report Cohen's κ; resolve disagreements to the lower level. Only AI_3 and AI_4 count toward P6.

Coding rules (from calibration cases in `ai-attribution/`):

1. Code discoveries documented in a publication or preprint; interview, blog and social-media statements are claims, not coded.
2. AI ranking candidates within a human-chosen target space: maximum AI_2.
3. Speed claims are not evidence; only measured SCT or Q_t.
4. Record source quality; only primary sources support levels above AI_1.
5. Formalizing a known result is a capability event for A, not a discovery.
6. A machine-checked proof counts as replication only after humans confirm the formal statement matches the theorem.
7. Flag conflicts of interest between an AI coder and the coded system; human coding first.

Each discovery is also assigned a domain group—(a) mathematics and formal sciences, (b) computational sciences, (c) experimental sciences—and an importance grade (1 incremental, 2 recognized open problem, 3 central problem), by the same two-coder procedure. Events before the preregistration date are logged as baseline events.

7. Longevity Translation Index

Track progression of an intervention through

$$L_0 \to L_1 \to L_2 \to L_3 \to L_4 \to L_5 \to L_6 \to L_7 \to L_8$$

with L_0 = hypothesis, L_1 = in vitro, L_2 = animal, L_3 = large animal, L_4 = Phase I, L_5 = Phase II, L_6 = Phase III, L_7 = approval, L_8 = demonstrated clinical / mortality benefit.

The model should estimate transition hazards and transition times between levels, using historical baselines (e.g. Wong, Siah & Lo 2019), rather than treating a preclinical result as equivalent to a clinical outcome.

Requirements (also for P7, P8):

- **Unit:** LTI level, EMS and G_j belong to the pair *(intervention, indication)*, not to the intervention alone (e.g. semaglutide: L_8 for obesity and cardiovascular risk, L_2–L_3 for slowing ageing).
- **Composition bias:** compare within therapeutic area × modality strata and IND-filing-year cohorts; combine strata with fixed weights. Repurposed drugs form a separate stratum.
- **Censoring:** estimate time to approval with Kaplan–Meier and Cox models (covariates: area, modality); model phase progression $L_4 \to L_5 \to L_6 \to L_7$ as a multistate model with failure as a competing absorbing state.

8. Evidence Maturity Score

Suggested ordinal mapping:

- 0.05 — hypothesis / mechanistic speculation
- 0.15 — in vitro
- 0.25 — animal
- 0.40 — human observational
- 0.55 — small controlled human trial
- 0.70 — large randomized controlled trial
- 0.85 — independent replication
- 1.00 — demonstrated clinically meaningful endpoint / mortality benefit with replication

Values are provisional. EMS is intervention-level (see Section 3).

9. Longevity Escape Velocity proxy

Escape condition. For an individual with population-average risk, remaining healthy life is $h(t) = HALE_{a(t)}(t)$ with $da/dt = 1$, so

$$\frac{dh}{dt} = \partial_t HALE_x + \partial_x HALE_x.$$

Escape velocity, $dh/dt \geq 0$, requires $\partial_t HALE_x \geq -\partial_x HALE_x$. The right-hand side is usually below 1 at older ages, so the v0.3.1 rule ($\partial_t HALE_x \geq 1$) was stricter than the escape condition.

Normalized period definition (framework default). With $HALE_x(t)$ the period healthy-life expectancy at age x (default $x = 65$) in year t:

$$LEV_x(t) = \frac{\partial_t \, HALE_x(t)}{-\partial_x \, HALE_x(t)}.$$

Interpretation:

- $LEV_x < 0$: healthy-life expectancy at age x is falling over calendar time.
- $0 < LEV_x < 1$: medicine offsets a fraction LEV_x of the healthy life lost per year of ageing at age x.
- $LEV_x \geq 1$: operational definition of longevity escape velocity; equivalent to $dh/dt \geq 0$ for an individual with population-average risk.

Also report the numerator $\partial_t HALE_x$ alone (raw frontier speed). Estimate $\partial_x HALE_x$ by finite differences across adjacent ages in the same period table; interpolate five-year age groups to single years with a method fixed at preregistration; report intervals.

Limits. The period proxy assumes current age-specific rates persist and does not describe individuals whose risk departs from the population's.

Reference, not an estimate. Record period life expectancy at birth has risen by about 0.25 years per year since 1840 (Oeppen & Vaupel 2002). This is a different series from $HALE_{65}$ and does not estimate LEV_{65}; the proxy's current value must be estimated from fixed-age HALE series.

This is a forecasting construct, not an accepted clinical metric.

10. Technology Survival Ladder

For milestone therapy class j—a pair (intervention class, indication)—define the time-dependent arrival hazard

$$\lambda_j(t) = \lambda_{0,j} \left[F_t^{sys}\right]^{\alpha_j} G_j(t)$$

where:

- $\lambda_{0,j}$: baseline hazard in 2026;
- α_j: sensitivity of class j to system-wide progress;
- $G_j(t)$: readiness factor from the class's current LTI level and historical transition probabilities, updated with EMS-weighted evidence.

Cumulative arrival probability from 2026 ($t = 0$) to time T:

$$P_j(T) = 1 - \exp\left(-\int_0^T \lambda_j(t)\,dt\right)$$

Population impact is $P_j(T) \cdot DAI_j$.

A person's "technology survival ladder" is the sequence of future medical milestones that become reachable while the individual remains alive and sufficiently healthy to benefit.

It is not a personal mortality calculator.

11. Monte Carlo forecasting

Future versions should:

1. Specify distributions for annual growth and slowdown in each subsystem.
2. Sample the elasticity of substitution σ (Section 2.2) and the input elasticity η (Section 2.4).
3. Sample bottlenecks and discontinuous breakthroughs.
4. Simulate at least 10^5 trajectories.
5. Report medians and 10/50/90% intervals.
6. Score forecasts retrospectively using Brier scores and calibration plots.
7. Update priors at fixed intervals rather than after emotionally salient news.

12. Falsification criteria

v0.3.3 is **not** the preregistration: it fixes the form of each prediction and the threshold rule. The preregistration is the frozen, timestamped output of that rule (planned: `forecasts/preregistration-2026.json`), made before any data from the tested windows are examined. Each window starts on the freeze date; earlier data are used only for baselines. Window labels below are nominal, assuming a freeze in late 2026.

Thresholds below are **placeholders**. Before preregistration, each is replaced by the larger of (1) the value exceeded with at most 5% probability under the historical regime, from the statistic's variability over past windows of equal length, and (2) the minimum scientifically or clinically relevant effect.

Multiplicity. For "at least k of m" predictions (P1, P3, P5, P7, P8), the 5% rule applies to the whole statement: simulate the probability that at least k of the m preregistered domains/strata pass under the historical regime, or use a Holm correction across strata where simulation is infeasible. Domain and stratum lists (hence m) are fixed at preregistration. All predictions are reported regardless of outcome.

Power. Before freezing, simulate each prediction under (a) the historical regime and (b) a minimally relevant acceleration, and publish the power. Predictions with power below 50% (likely P2 and P4, which estimate a change in a growth rate over a short window) get a longer window or are marked exploratory.

Report 50/80/90/95% intervals; the **90% interval** is the preregistered decision criterion.

#	Prediction	Window	Supports the hypothesis if…
P1	Cross-domain acceleration	2026–2031	$TAR_i > 1.5$, lower 90% bound > 1, in at least 3 HTAB domains
P2	Rising growth of capabilities and efficiency	2026–2031	$dg_O/dt > 0$ and $dg_\Pi/dt > 0$ (Section 2.4), each with 90% interval excluding 0
P3	Research productivity reverses	2026–2036	Research productivity against R^{total} (Section 4.1, Bloom et al. 2020 sense) rises in at least 2 of their domains
P4	Autonomy accelerates	2026–2031	METR 50% time-horizon doubling time over the window shorter than over the pre-freeze baseline (90% interval of the ratio below 1), and non-software components of A rising
P5	Faster experimental loops	2026–2031	Median SCT falls $\geq$ 50% in at least 2 self-driving-lab domains, with no fall in Q_t
P6	Replicated AI discoveries	2026–2031	Annual independently replicated AI_3/AI_4 discoveries (Section 6.1), relative to the baseline frozen at preregistration, at least double and exceed a preregistered absolute minimum, in raw and importance-weighted counts; and the experimental-sciences count rises (a rise confined to mathematics is partial support)

#	Prediction	Window	Supports the hypothesis if…
P7	Faster translation	2026–2036	Median IND-to-approval time (survival analysis, within area × modality strata, excluding repurposed drugs) falls $\geq$ 20% vs 2015–2025 IND cohorts in at least one therapeutic area
P8	Better clinical success	2026–2036	Phase I-to-approval probability (multistate model, within strata, excluding repurposed drugs) improves $\geq$ 30% vs Wong et al. 2019 in at least one area

Exploratory statistic. Debut rate of cost decline across successive AI performance levels (Epoch AI 2026): rising debut rates would indicate acceleration in AI efficiency; within-level slowdown is an ordinary learning curve. Reported with P2/P4, not counted in the multiplicity correction.

The Saka Law should be weakened or rejected as a useful forecasting hypothesis if, by the end of the relevant window:

- TAR_i is not distinguishable from 1 in a majority of HTAB domains;
- AI capability and autonomy improve but SCT, Q_t and replication do not (P5, P6 fail);
- biological translation times and success probabilities remain statistically unchanged (P7, P8 fail);
- $LEV_{65}(t)$ in the largest high-income populations shows no increase over its own 2000–2025 trend through 2046;
- persistent physical, economic or regulatory bottlenecks dominate the feedback loop.

Success of P1–P6 with failure of P7, P8 and the LEV criterion would support acceleration in computation and discovery but not its extension to biomedicine.

The model is therefore designed to permit a negative result.

13. Candidate data sources

Sources are proposals; each must be fixed, with version and access date, at preregistration. Access terms change and should be re-checked when data are pulled.

13.1 State vector

Component	Candidate sources	Access
A	METR time-horizon data; Epoch AI Benchmarking Hub (FrontierMath and others); formal-proof benchmarks (e.g. miniF2F, PutnamBench); Epoch AI, "The plunging price of thought" (cost of a fixed capability level, 2023–)	Open
C	Epoch AI Data Hub (notable models, ML hardware, compute stock); TOP500	Open
E	Ember electricity data; IEA data-centre estimates; LBNL "Queued Up" interconnection-queue data	Ember and LBNL open; IEA partly
R	No central database; to be extracted from self-driving-lab publications in fixed domains	**To build**
B	NHGRI sequencing-cost series; ClinicalTrials.gov for modality classes in human trials	Open
D	Published cohort sizes (e.g. UK Biobank, All of Us, FinnGen)	Sizes open; data controlled
S	Jonathan McDowell's GCAT launch catalogue; published cost-per-kg estimates	Open
M	Industry reports on wafer starts (e.g. SEMI, foundry filings); cell- and gene-therapy capacity surveys	Mostly paid
L	AACT (ClinicalTrials.gov as a relational database); Drugs@FDA; FDA novel approvals	Open

13.2 HTAB baseline

- Santa Fe Institute Performance Curve Database (the data behind Nagy et al. 2013 and Farmer & Lafond 2016).
- Our World in Data technology series (transistors, solar, batteries, sequencing), with original sources.
- Introduction dates and cumulative production for each series, for maturity-aligned TAR.
- Power-electronics cost and efficiency (industry and IEA sources); cost of a fixed AI capability level (Epoch AI). Caveats for the latter: three years of data, benchmarks may be targeted in training, prices include margins, results range 43–58% per quarter by averaging method.
- Stagnating domains: FAOSTAT crop yields; BLS construction productivity; Eroom's-law

series (Scannell et al. 2012).
- Bloom et al. (2020) replication package (openICPSR), needed for P3.

13.3 Longevity and translation

- Human Mortality Database (period life tables by single year of age).
- HALE by age: IHME Global Burden of Disease; WHO Global Health Observatory.
- Clinical transition probabilities: Wong et al. (2019) used proprietary data; an open replication must be rebuilt from AACT linked to Drugs@FDA.

13.4 To be built by the project

- Scientific Cycle Time and Q_t in 2–3 fixed self-driving-lab domains, extracted from methods sections.
- AI_0–AI_4 attribution: candidate discoveries from OpenAlex/Crossref metadata (including CRediT roles where present), coded by two independent coders. If an AI system is used as one coder, the second must be human.